

**Name:** Team Timbuktu

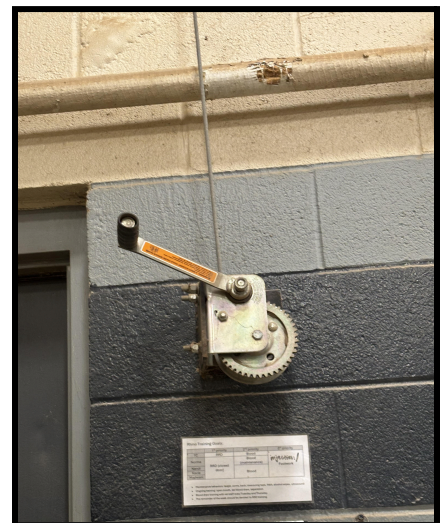
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## ***Element B: Documentation and Analysis of Prior Design Attempts***

### **Prior Attempt 1: Crank at Nashville Zoo for Rhino Holding Area**

The crank at the Nashville Zoo was one of the prior attempts we saw in person. Observing the crank in action, it was evident how it made moving the toys up and down significantly easier. This was primarily because the gears were doing most of the heavy lifting, reducing the effort required to operate the mechanism. The efficiency and simplicity of this design stood out to us, highlighting its potential for various applications. A design like this would be ideal for our needs, as it offers a practical solution to our problem. The gears' contribution to the overall function means that even with minimal physical effort, the toys can be moved smoothly and effectively. This could be particularly beneficial in settings where ease of use and reliability are crucial factors.

However, we need to consider our budget, which is limited to \$200. While the crank design from the zoo is a fantastic model to follow, we must ensure that we can replicate or adapt it within our financial constraints. This may involve sourcing cost-effective materials or simplifying certain design aspects without compromising its core functionality. With careful planning and creativity, we can develop a solution that meets our requirements and stays within our budget.



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### **Prior Attempt 2: Hanger at Nashville Zoo for Giraffe Holding Area**

The hanger in the giraffe holding area is cleverly constructed from a combination of a rope and an anchor, which work together to maintain the desired height of the toy securely in place. This design is an excellent candidate for our project due to its simplicity and ease of use; it only requires tying the rope around a hook, eliminating the need for complicated mechanisms or extensive adjustments. Such a straightforward approach ensures the design is both practical and user-friendly, essential qualities for our considerations. The simplicity of this system is a significant advantage, particularly for caretakers who need to interact with it regularly. A reliable and easy-to-manage mechanism allows for quick adjustments to the toy's height, promoting an engaging environment for the animals with minimal effort. Additionally, the lack of complex components reduces potential maintenance issues, providing confidence in its safety and reliability. However, we must note that the overall size of this design is quite large, which may render it unsuitable in its current form for our specific application. The large scale could create challenges related to installation, spatial constraints, and even aesthetics within the environment. A significant size might restrict the functionality we aim to achieve within the designated space for the elands. To effectively adapt and incorporate this concept into our project, we will need to scale it down considerably to align with our design requirements and constraints. This scaling process will involve re-evaluating the dimensions and materials used in the hanger, ensuring it retains its essential function while fitting seamlessly within our project's parameters. A more compact design will not only fit better in the area but also enhance the aesthetics, making it less intrusive. We must also be mindful to maintain the mechanism's stability and strength while downsizing. By choosing durable yet lightweight materials, we can ensure the system remains effective in securely holding the toy at the intended height, allowing for safe and engaging play for the elands. In summary, while the

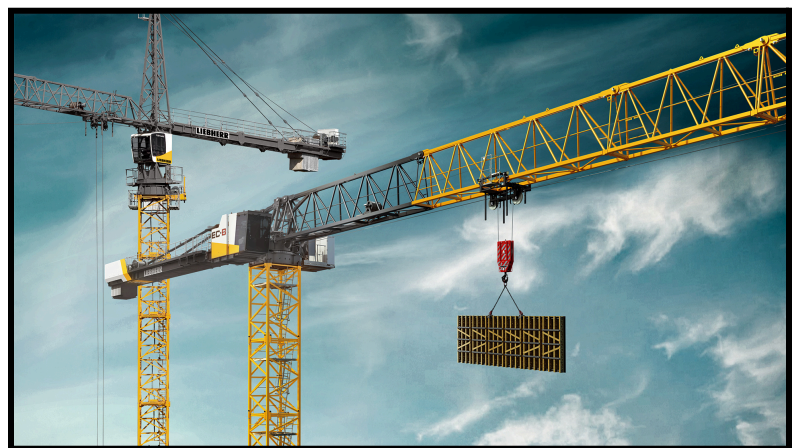
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hanger designed for the giraffe holding area exemplifies simplicity and practicality, adapting it for our specific needs will require careful adjustments. By scaling down the design and refining its dimensions, we will create a functional and user-friendly solution that effectively maintains the desired height of the toy, ultimately enhancing the elands' play experience and well-being.

### **Prior Attempt 3 - Industrial Tower Crane**

Some of our designs resemble tower cranes in both their aesthetic and functional characteristics, bringing a sense of industrial elegance to our applications. However, there is a significant difference that sets our designs apart: size. Tower cranes are engineered to lift thousands of pounds high above city skylines and across extensive construction sites, whereas our cranes are specifically designed to lift a smaller load of just 100 pounds to a height of 10 feet. This distinction is critical because it allows us to tailor our design to meet the unique needs of the Elands pen. Since the pen for the Elands is relatively compact, our design doesn't require rotation, which is a fundamental feature of traditional tower cranes. By eliminating the need for this rotational movement, we can concentrate our efforts on enhancing the strength and stability of the hanger itself. The reduction in complexity not only streamlines our design but also minimizes potential points of failure, leading to a more reliable and user-friendly lifting mechanism.



Focusing on these factors is crucial because the primary goal is to ensure that our system

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can lift the specified 100 pounds effectively and with minimal risk. The simplicity inherent in our design reduces maintenance requirements over time, which is particularly advantageous for long-term operation. This effort makes it easier to maintain the lifting mechanism and reduces downtime that could affect the functionality of the Elands pen. Moreover, without the constraints of a rotating mechanism, we are afforded greater freedom in selecting materials and construction methods.

We can prioritize using high-strength materials that enhance durability, ensuring that the hanger can withstand the forces exerted during operation. This targeted approach to material selection allows us to build a lifting mechanism that is not only capable of handling the necessary loads but also upholding the highest safety standards. The choice of materials is particularly important in our design philosophy. By evaluating options that are robust yet lightweight, we can maintain the integrity of the hanger while optimizing its performance. This careful consideration extends to every aspect of the construction process, from the initial design phase through to final assembly. In essence, the result is a lifting mechanism that is not only strong and dependable but also tailored to meet the specific requirements of the Elands pen.

By understanding and addressing these unique needs, we have developed a solution that exemplifies our commitment to safety, efficiency, and reliability. We believe that by honing in on these specific design parameters, we can deliver a lifting solution that exceeds expectations while ensuring the structural integrity and safety of the environment in which it operates.



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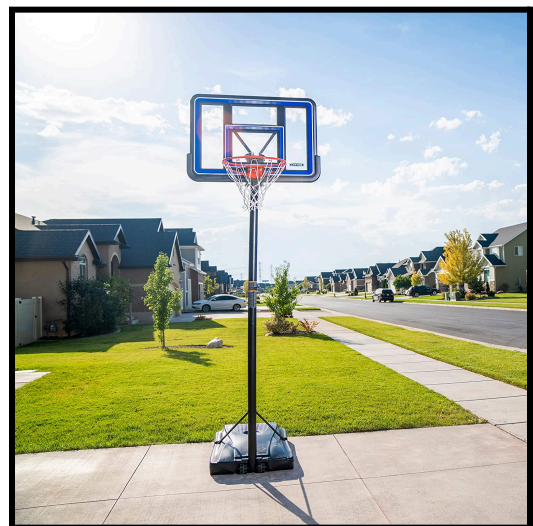
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### **Prior Attempt 4: Basketball Hoop**

This design features a base that is quite similar to something we will be using, and it also mirrors the size of the Enrichment Toy Hanger that we are planning to build. The significance of the hoop's base cannot be overstated in this context, as the hanger will need to have the capability to lift a toy weighing 125 pounds to a height of 10 feet in the air. The design of the hoop's base is critical since it is responsible for ensuring that the hoop can stand upright, as it is firmly anchored to the ground.

Given that the toy's considerable weight necessitates a sturdy and reliable support system, the base plays a pivotal role in the overall stability and functionality of the hanger. The engineering behind the base is designed to provide a secure foundation that can handle the dynamic forces exerted by the heavy toy. This is particularly important because the hanger must remain stable and upright while lifting and supporting the toy at such a significant height.

The weighted base of the hoop allows it to maintain a straight and stable position, which is crucial for the effective operation of the hanger. The design ensures that the base remains grounded, preventing any tilting or toppling that could compromise the safety and effectiveness of the hanger. By anchoring the hoop securely to the ground, the base provides the necessary support to handle the substantial weight of the toy, ensuring that it can be lifted and suspended at the desired height without any issues.



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### **Prior Attempt 5: Fishing Pole**

The fishing pole is an invaluable addition to our design, as it plays a crucial role in maintaining the chain at an appropriate height for the elands. By utilizing the fishing pole mechanism, we can effectively stop the line at a specific point, ensuring that the hook remains steady and secure. This innovative feature not only allows us to suspend the toy in a fixed position above the ground but also provides the flexibility to adjust the height as needed, accommodating the varying needs of the elands. Incorporating this adjustable system into our design significantly enhances the stability and functionality of the toy. For example, if we notice that the elands are interacting with the toy differently at various heights—perhaps showing more interest or

engagement when it is lower—we can easily modify its position to better suit their preferences. This adaptability is key in fostering positive interactions and encouraging the elands to explore and play. So, whether they enjoy reaching for the toy or prefer it closer to the ground, the fishing pole allows us to accommodate these dynamic preferences. Moreover,



this adjustable mechanism opens up the possibility for multiple play scenarios, which can further promote physical activity and mental engagement. By varying the toy's height, we can create different challenges for the elands, stimulating their curiosity and encouraging them to exercise. This added versatility enhances the overall experience for the animals, making their environment more enriching and interactive.

Additionally, we can explore the use of lightweight materials for the fishing pole to ensure that it remains easy for users to operate. Whether it's caretakers or visitors to the habitat, a lightweight pole will make it more convenient to adjust the toy's position without

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straining or requiring excessive effort. By prioritizing usability in our design, we not only achieve the desired height and function but also ensure that the mechanism is user-friendly for anyone interacting with it. Overall, the fishing pole concept represents an innovative solution for maintaining the toy's position while accommodating the dynamic interactions of the elands. By blending functionality with ease of use, we are contributing to a more engaging and responsive play environment, ultimately enhancing the well-being and enjoyment of the Elands. This thoughtful design approach underscores our commitment to creating enriching experiences that cater to the specific behaviors and needs of the eland.